



WIND AND WAVES

Winds blowing over the sea whip the surface into waves. The stronger the wind, the bigger are the waves. Waves also grow as they travel, so the highest waves are those that have traveled great distances over broad oceans. They have great destructive power, especially on exposed coasts where breaking waves can splinter solid rock. Out at the sea, they cause less damage, although rare rogue waves may sink even big ships.

◀ WAVE ENERGY

Waves are caused by friction or drag as the wind sweeps over the ocean. The wind often blows the spray forward off the wave crests. However, the water within the wave does not move forward with it. Each water particle moves in a circle, rolling forward and then back as the wave passes. This action transfers the energy forward, which can become destructive when the wave finally breaks on the shore.



WAVE DEVELOPMENT

RIPPLES

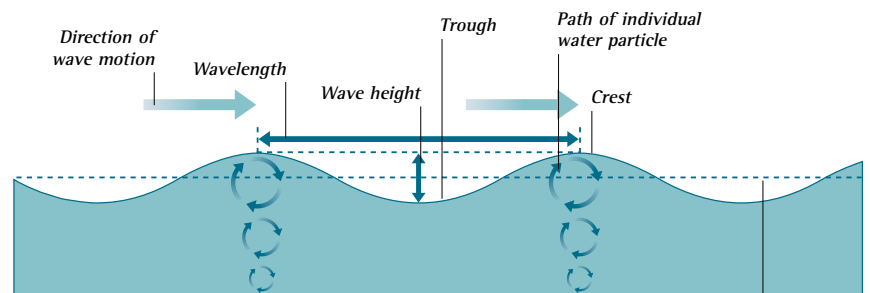
Ocean waves begin their lives as ripples on a flat sea, just like the ripples that are blown across a pond by the breeze. They are very small and close together, and from a distance they make the water surface shimmer without giving any impression of wave movement. If the wind keeps blowing, however, the ripples can grow into bigger waves.

CHOP

Ripples grow bigger and develop into a confused choppy sea, with waves up to 20 in (50 cm) high. The distance between the crests can vary from 10 ft (3 m) to 40 ft (12 m). They interact with each other to create a chaotic effect. These disordered waves are typical of areas where the waves are being whipped up by strong winds.

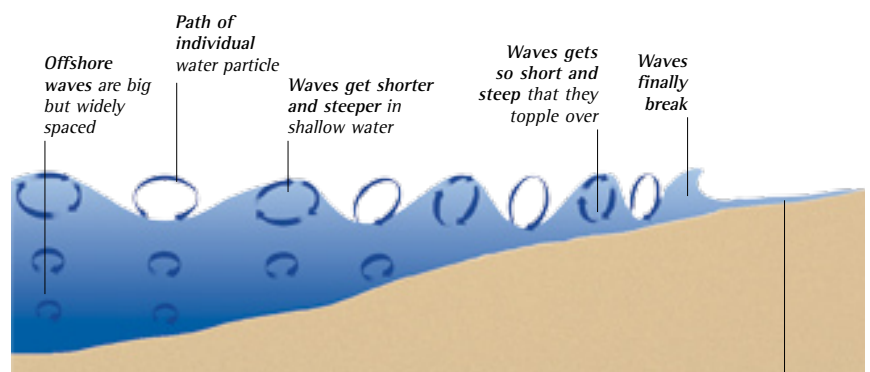
SWELL

The disordered chop of the sea becomes a regular series of waves called a swell. Though the distance between crests is much more predictable, the waves can grow higher if the wind is still blowing and the waves are able to travel over a big ocean. The swell seen here is breaking on the shore.



▲ LENGTH AND HEIGHT

Some waves are chaotic, especially if they are formed by strong winds or opposing currents. Most waves, however, develop into a regular succession of crests and troughs. The wave height is measured from trough to crest. The distance between each crest is called the wavelength, while the time delay between them is known as the period.



▲ BREAKING WAVES

As the waves approach the shore, the water becomes shallower. The waves become shorter and steeper as they are slowed down due to the interaction with the seabed. As each wave gets taller, it becomes less stable, and it finally topples forward to break on the shore. The steeper the shore, the more dramatically the waves break, releasing clouds of spray and driving a mass of water up the shore.